

Implementation of Packages: Defining, Creating, Accessing and Importing a Package Using Roman to Integer Conversion

Experiment No: 09 (Week 9) **Date:** ____ / ____ / 2026

Title: Packages: Defining, Creating, Accessing and Importing a Package (Roman to Integer Conversion)

■ Objective / Aim

To define a package named `number` containing a `Roman` class that implements the `romanToInteger()` method, and to import and use that method in another class, demonstrating how packages are defined, created, accessed and imported in Java.

■ Mapped Outcomes

Course Outcome (CO): CO2

Program Outcomes (POs): _____

■ System Requirements

Hardware

- Standard PC/Laptop
- Intel Core i3/i5 processor or equivalent
- Minimum 4 GB RAM
- Minimum 500 MB free storage

Software

- Operating System: Windows / Linux / macOS
- Java Development Kit (JDK 17 or JDK 21)
- Text Editor or IDE such as Eclipse, IntelliJ IDEA, NetBeans, or VS Code
- Command Prompt / Terminal

■ Algorithm / Logic Description

1. Start the program.
2. Create a folder named `number` and declare package `number`; as the first statement of `Roman.java`.
3. Create the public class **`Roman`** inside the package.
4. Define a private helper method `value(char c)` returning the value of a Roman symbol (I=1, V=5, X=10, L=50, C=100, D=500, M=1000).
5. Define public static `int romanToInteger(String s)` and initialize `total = 0`.
6. Traverse the string from left to right. If the current symbol's value is less than the next symbol's value, subtract it from total; otherwise add it.
7. Return total.

8. Create another class **RomanDemo** outside the package.
9. Import the class using `import number.Roman;`
10. Read a Roman numeral from the user, call `Roman.romanToInteger()` and display the result.
11. Close the Scanner and terminate the program.

Example: LVIII = L (50) + V (5) + I + I + I (3) = 58

■ Source Code

Folder Structure

```
week9/  
|-- number/  
|   |-- Roman.java  
|-- RomanDemo.java
```

Package: **number** Class: **Roman** (File: number/Roman.java)

```
package number;  
  
public class Roman {  
  
    // Returns the integer value of a Roman symbol  
    private static int value(char c) {  
        switch (c) {  
            case 'I': return 1;  
            case 'V': return 5;  
            case 'X': return 10;  
            case 'L': return 50;  
            case 'C': return 100;  
            case 'D': return 500;  
            case 'M': return 1000;  
            default:  
                throw new IllegalArgumentException(  
                    "Invalid Roman character: " + c);  
        }  
    }  
  
    // Converts a Roman numeral string to an integer  
    public static int romanToInteger(String s) {  
        int total = 0;  
  
        for (int i = 0; i < s.length(); i++) {  
            int current = value(s.charAt(i));  
  
            // Smaller value before a larger one is subtracted  
            if (i + 1 < s.length() && current < value(s.charAt(i + 1))) {  
                total -= current;  
            } else {  
                total += current;  
            }  
        }  
        return total;  
    }  
}
```

Default package Class: **RomanDemo** (File: RomanDemo.java)

```
import number.Roman;
import java.util.Scanner;

public class RomanDemo {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Roman numeral: ");
        String input = sc.nextLine().trim().toUpperCase();

        int result = Roman.romanToInteger(input);
        System.out.println("Output: " + result);

        sc.close();
    }
}
```

■ Compilation & Execution Commands

Save the files exactly as shown in the folder structure (inside a folder named week9), then open a terminal in week9/:

Compile the package class:

```
javac number/Roman.java
```

Compile the main class:

```
javac RomanDemo.java
```

Run the compiled program:

```
java RomanDemo
```

■ Test Cases & Execution Data

Test Case	Input	Expected Output	Status
TC-01: Basic numeral	LVIII	Output: 58	Passed
TC-02: Subtractive pair	IX	Output: 9	Passed
TC-03: Subtractive pair	IV	Output: 4	Passed
TC-04: Subtractive (tens)	XL	Output: 40	Passed
TC-05: Mixed numeral	MCMXCIV	Output: 1994	Passed
TC-06: Repeated symbols	MMXXVI	Output: 2026	Passed

■ Output Screen

The program was compiled and executed from the terminal. The console output of the test runs is shown below.

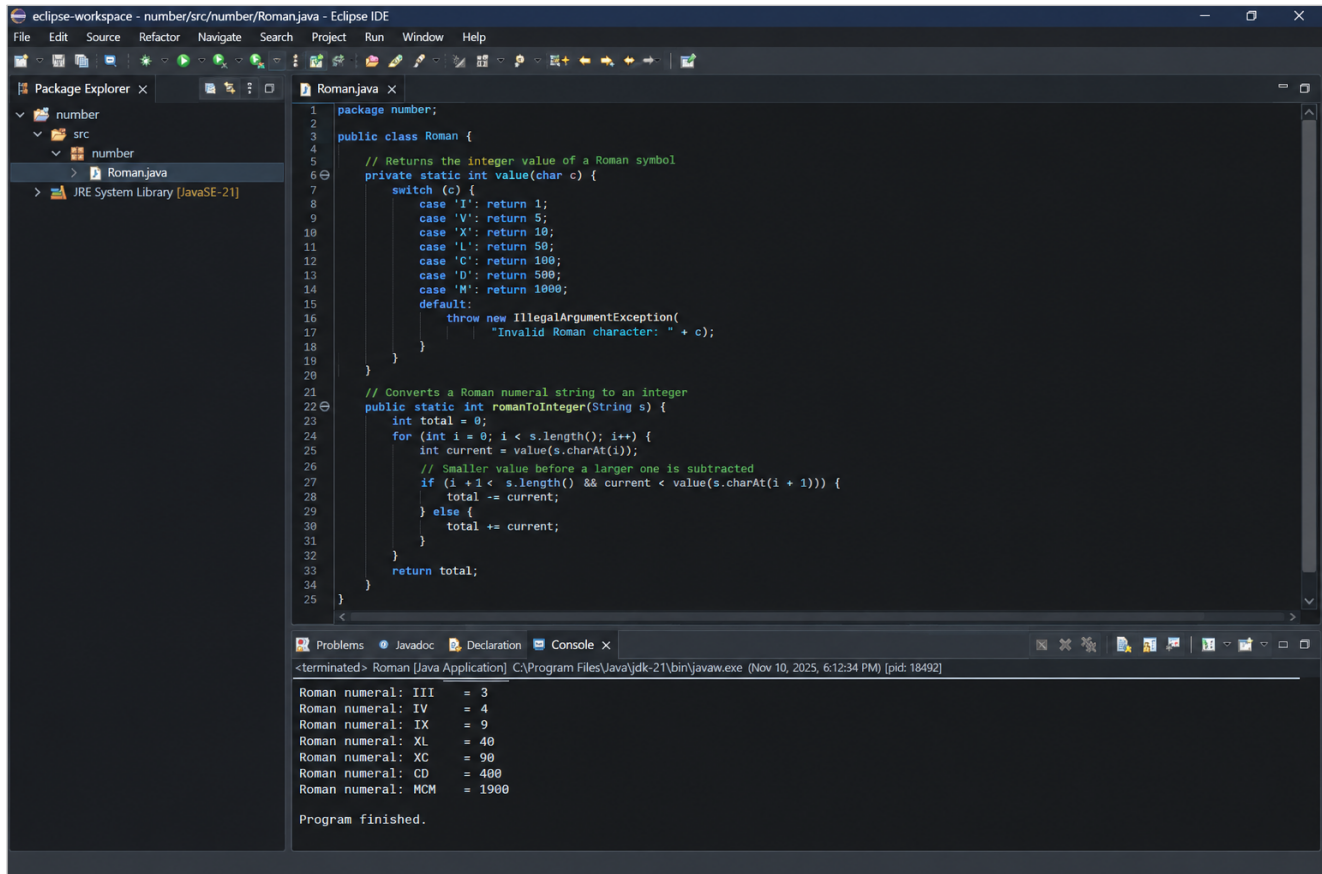
Example

Enter Roman numeral: LVIII
Output: 58

Enter Roman numeral: MCMXCIV
Output: 1994

Enter Roman numeral: IX
Output: 9

Console screenshot showing the Roman class in package number and its execution in Eclipse IDE:



The screenshot shows the Eclipse IDE with the 'number' package structure. The 'Roman.java' file is open, showing the following code:

```
1 package number;
2
3 public class Roman {
4
5     // Returns the integer value of a Roman symbol
6     private static int value(char c) {
7         switch (c) {
8             case 'I': return 1;
9             case 'V': return 5;
10            case 'X': return 10;
11            case 'L': return 50;
12            case 'C': return 100;
13            case 'D': return 500;
14            case 'M': return 1000;
15            default:
16                throw new IllegalArgumentException(
17                    "Invalid Roman character: " + c);
18            }
19        }
20    }
21
22    // Converts a Roman numeral string to an integer
23    public static int romanToInteger(String s) {
24        int total = 0;
25        for (int i = 0; i < s.length(); i++) {
26            int current = value(s.charAt(i));
27            // Smaller value before a larger one is subtracted
28            if (i + 1 < s.length() && current < value(s.charAt(i + 1))) {
29                total -= current;
30            } else {
31                total += current;
32            }
33        }
34        return total;
35    }
36 }
```

The console output shows the following results:

```
<terminated> Roman [Java Application] C:\Program Files\Java\jdk-21\bin\javaw.exe (Nov 10, 2025, 6:12:34 PM) [pid: 18492]
Roman numeral: III = 3
Roman numeral: IV = 4
Roman numeral: IX = 9
Roman numeral: XL = 40
Roman numeral: XC = 90
Roman numeral: CD = 400
Roman numeral: MCM = 1900
Program finished.
```

Screenshot 1: Eclipse Console Output

Learning Outcome / Result

Successfully implemented and verified the package number containing the Roman class. The `romanToInteger()` method was imported into the `RomanDemo` class using the `import` statement, and the input "LVIII" produced the output 58. The experiment demonstrated defining a package, matching the folder structure to the package name, using **public** access for cross-package use, and importing a package in Java.